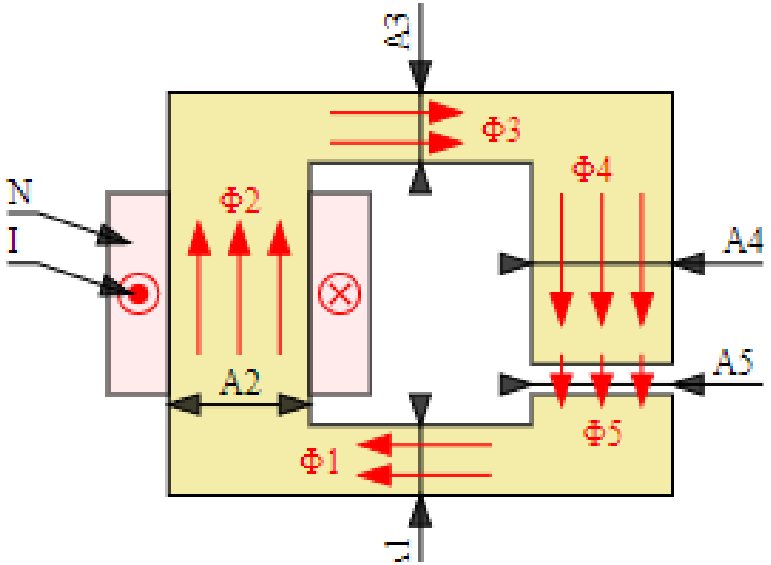
ECE 330

Chapter 4

Magnetic circuits, Mutual Inductance, and Transformers



Prof.Olga Mironenko

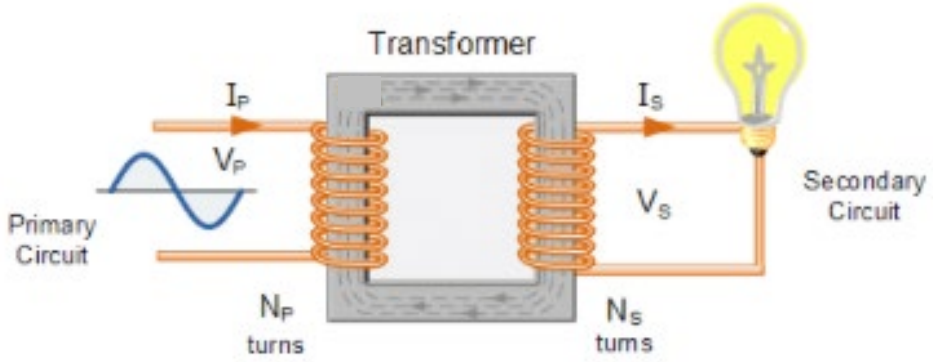
University of Illinois at Urbana-Champaign

Fall 2026

Learning Objectives: you will be able to

- Analyze **ideal transformers,** and determine voltage, current and impedance relationships between primary and secondary sides.
- Use **Maxwell's equations** to analyze an inductor, and to develop **magnetic equivalent circuits.**
- Model magnetic systems using **magnetic equivalent circuits**, including **reluctance** and **inductance**.
- Evaluate airgap effects, accounting for **airgap reluctance** and **fringing effects.**
- Analyzed **coupled coils**, using **mutual inductance** and **the dot convention.**

Ideal Transformers



Why Edison lost the War of Currents

Moving Impedances Across a Transformer

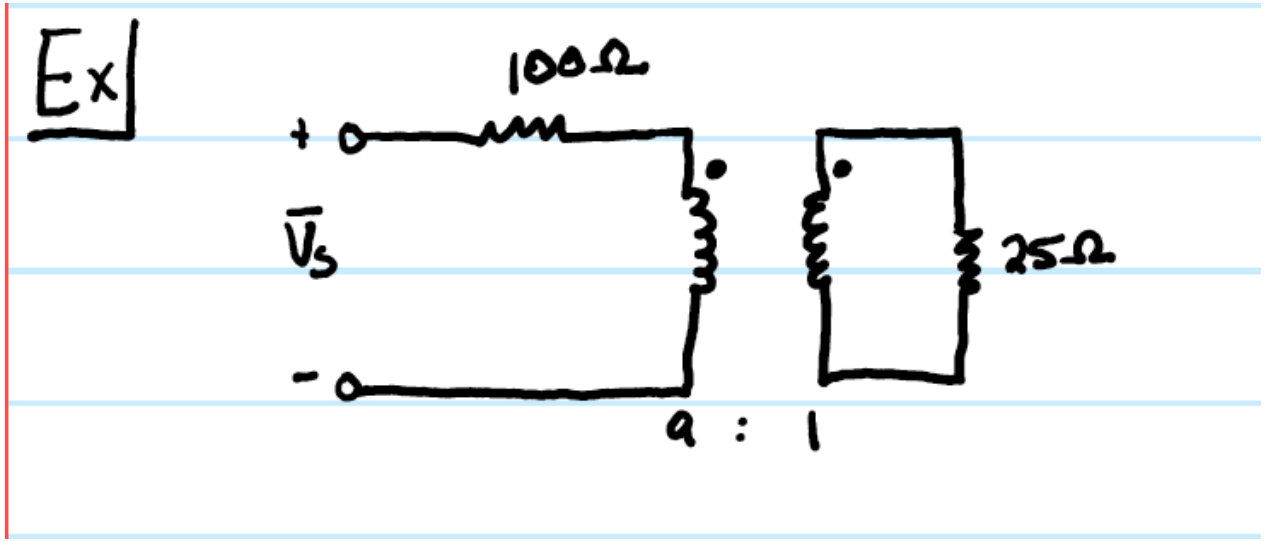
Maximum Power Transfer

Maximum Power Transfer (Continued)

- Can show for maximum power dissipated at the load
  - The load impedance is the complex conjugate of the line impedance
  - For a fixed ratio between the resistive and reactive terms of the load impedance, the magnitude of the load impedance is the same as the line impedance

Maximum Power Transfer with Transformer

Example 1



What is *a* for maximum power transfer?

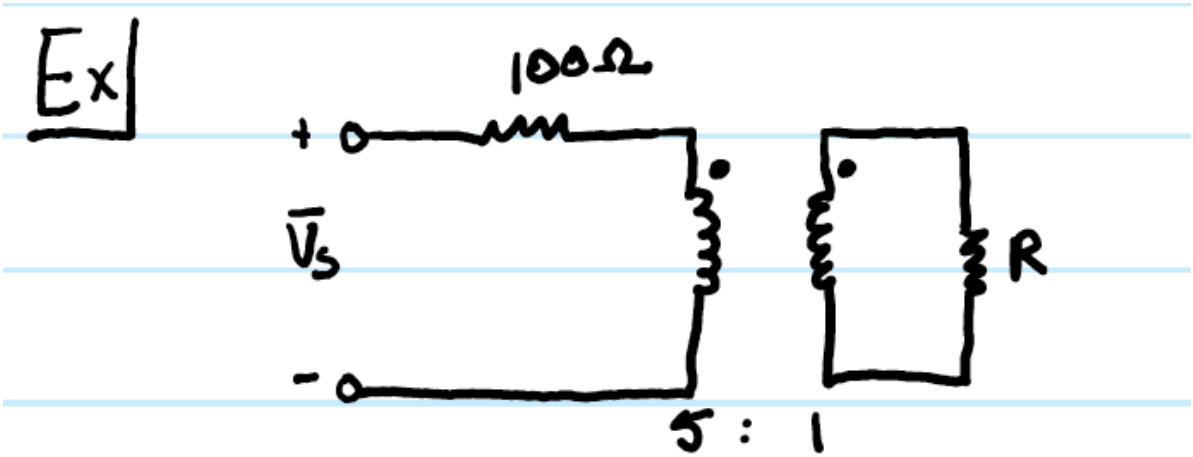
a) 1

b) 2

c) 3

d) 4

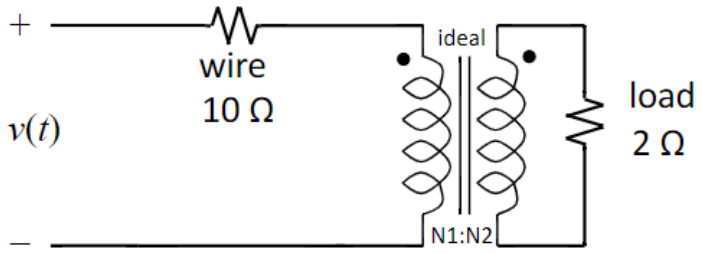
Example 2



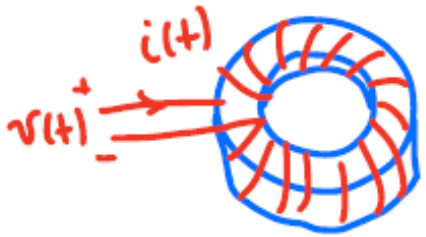
What is R for maximum power transfer?

Example 3 (Pr 4d, Spring 2021, M1)

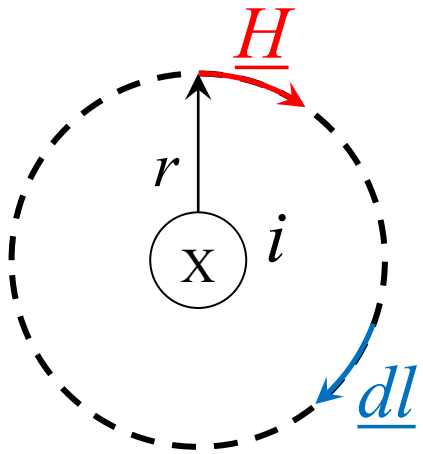
d) The purpose of our transformer is to drive a 2 Ω load from a source that is connected through a 10 Ω wire, as shown below. If v(t) = cos(t) V and we model the transformer as ideal, what is the average power delivered to the load?



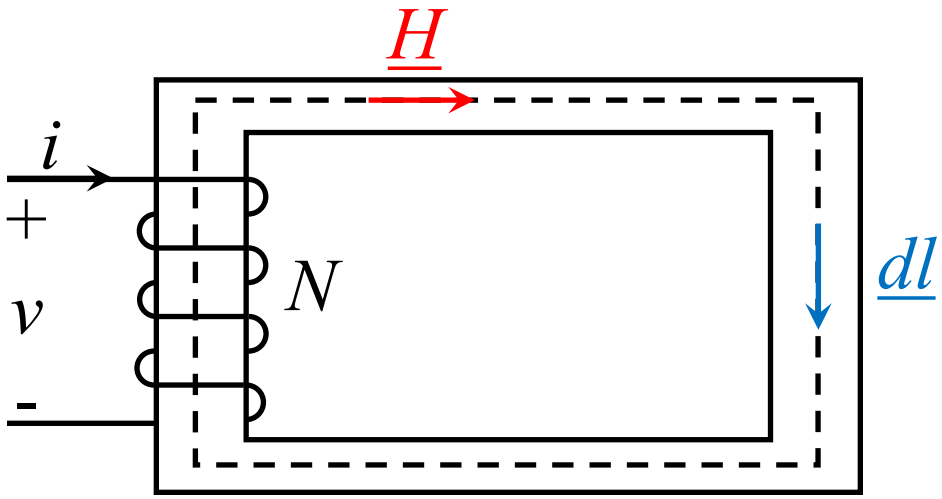
Electromagnetics



Electromagnetics



Electromagnetics



Magnetic Equivalent Circuit

Magnetic **Electrical**

What about the *E*?

Inductance

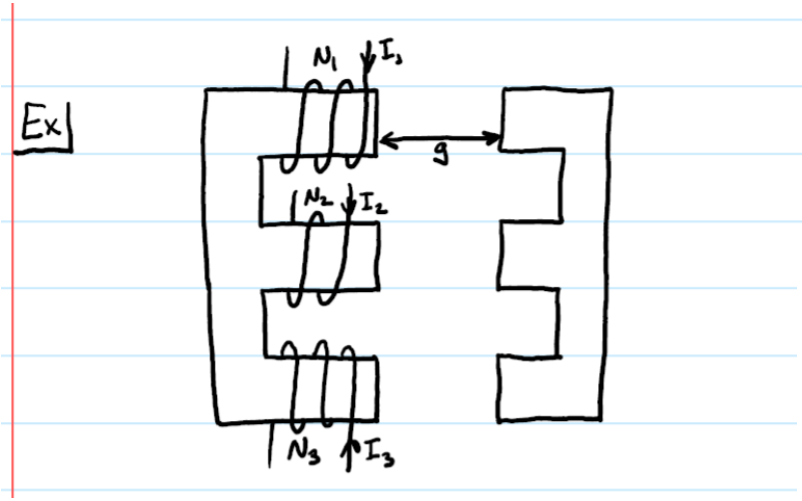
Iron Core with Air Gap

Iron Core with Air Gap

Faraday's Law

Fringing Effects

Example 4



Given: $N_1 = 100$ $I_1 = 25A$ $A_g = 4cm^2$

$N_2 = 50$ $I_2 = 10A$ $\mu = \infty$

$N_3 = 100$ $I_3 = 15A$ $g = 0.1cm$

Where should the positive terminals be placed on the MMF sources?

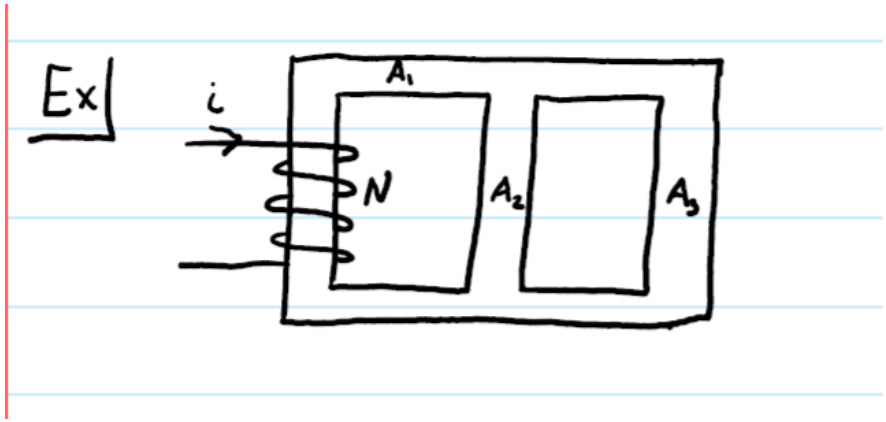
a) right, left, right

b) left, left, right

c) right, right, left

d) right, right, right

Example 5



Coupled Coils: ideal case

Dot convention

Dot convention

- Step 1) Select a coil and put a dot on one terminal- arbitrarily (your choice)
- Step 2) Mark the dotted terminal with + voltage polarity and inject current into first dot
- Step 3) Use RHR to determine flux direction
- Step 4) Determine current direction in the second winding which will throw flux in the same direction as the first winding
- Step 5) Put the dot in the second winding, where current enters the terminal
- Step 6) Put + voltage polarity on the second dotted terminal

Coupled Coils: practical case

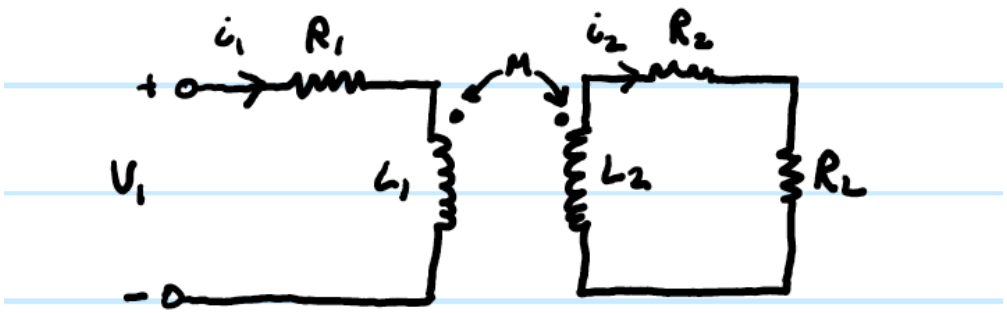
Ideal:

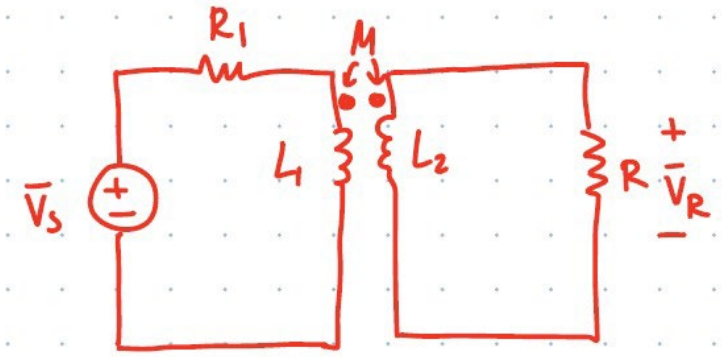
Practical:

Coefficient of Coupling

Example 6

Real Transformers





Given: ω, $\bar{V}_S$, R_1, R, L_1, L_2, M
Find: $\bar{V}_R$

- Real transformers have resistance in the wire, inductance from the core, and mutual inductance between the two windings.

Side note: Thevenin Equivalent (not on quizzes/exams)

